
Exercise Sheet 4:
Tensors in probability theory
Foundations of Neuro-Symbolic AI
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Server Diagnostics

Let there be three Boolean variables:

- X_0 - Whether there is high traffic.
- X_1 - Whether the server crashes.
- X_2 - Whether the crash alert gets triggered.

Consider a joint distribution of these variables defined by the uncertainty table:

<i>Traffic</i> x_0	<i>Crash</i> x_1	<i>Alert</i> x_2	$\mathbb{P}[X_0 = x_0, X_1 = x_1, X_2 = x_2]$
0	0	0	0.648
0	0	1	0.072
0	1	0	0.015
0	1	1	0.135
1	0	0	0.072
1	0	1	0.008
1	1	0	0.005
1	1	1	0.045

Task 1 (after lecture II.0-Distributions):

Align the uncertainties in an order-3 tensor $\mathbb{P}[X_0, X_1, X_2]$. Compute the contractions and normalizations providing:

- (i) The marginal distribution of the variables X_0 and X_1 , describing the joint distribution of *Traffic* and *Crash*.
- (ii) The conditional distribution of X_1 conditioned on X_0 , describing the distribution of *Crash* conditioned on *Traffic*.

Task 2 (after lecture II.0-Distributions):

Compute the probability of the event, that the server crashes whenever the traffic is high.

- (i) Find the propositional formula corresponding with the event.
- (ii) Contract the formula with the distribution to compute the events probability.

Task 3 (after lecture II.1-Independence):

Decide based on contractions of $\mathbb{P}[X_0, X_1, X_2]$:

- (i) The server crashes independently of the traffic.
- (ii) The server traffic and the crash alert are independent conditioned on the server crash.

Task 4 (after lecture II.1-Independence):

Represent the distribution as a Bayesian Network. Draw the underlying hypergraph and find the decorating conditional probability tensors.